

Karan Sidhu

karan02sidhu@gmail.com | 780-782-0840 | Edmonton, AB | [LinkedIn](#) | [GitHub](#)

Education

University of Alberta

Bachelor of Science | GPA: 3.3

Sept 2020 - Jun 2025

Double Major in Computer Science and Biology

Relevant Coursework: search algorithms, supervised learning, unsupervised learning, reinforcement learning, parsing/tagging texts, error analysis, text normalization, chatbots, storage architecture design, UML, software design patterns, software architecture, object-oriented programming

Skills

Programming: Python, Java, ReactJS, html5, CSS3, C

DBSM: SQL, MongoDB, Firebase, Neo4j Cypher (GraphQL)

AI/ML: Prompt engineering, fine-tuning ML models, relation extraction, chatbots

Software: GitHub, bitbucket, Huggingface, Docker, Android Studio, Linux command line and bash scripting

Experience

TMIC Wishart Node | Artificial Intelligence / Software Intern

May 2024 - Sep 2024

- Delivered weekly presentation project updates, and ensuring alignment on research goals.
- Enhanced knowledge graph construction through an AI pipeline that extracts data (triplets) from research articles using LLM's.
- Fine-tuned HuggingFace models, improving relation extraction accuracy by 14.6%.
- Engineered a cost-effective alternative to AppSignal for error and performance monitoring.
- Designed custom GPT chatbots to streamline and automate the grant application processes.

Projects

ChemFont Fact Finder | ([Fact Finder](#), [Project7](#), [TextMining Chatbot](#))

Jan 2024 - Sep 2024

- Designed and implemented a pipeline to extract triplets from research articles.
- Developed high-quality datasets that enhanced evaluation accuracy and model assessments.
- Engineered a system of bonus positives to refine evaluation metrics, addressing previously unaccounted valid predictions and improving model assessment reliability.
- Automated quantitative evaluation, increasing accuracy and consistency of model assessments.

AppSignalReplicate | ([App Signal Replicate](#))

Jul 2024 - Aug 2024

- Engineered a lightweight, free alternative to AppSignal for error tracking and performance monitoring across multi-server environments.
- Automated log analysis across servers via SSH with real-time email notifications for anomalies, ensuring proactive error detection.

Event QR Check-in App | ([QR Check-in App](#))

Jan 2024 - Jun 2024

- Designed and developed an Android app for event management using Android Studio and Firebase, enhancing event check-ins for attendees, organizers, and admins.
- Engineered QR code generation and scanning modules, integrating them with real-time database queries to support event workflows.
- Oversaw agile cycles and tracked tasks with GitHub Projects, ensuring delivery of milestones.

Portfolio Chatbot Website | ([Public Portfolio](#))

May 2025 - Jun 2025

- Built a modern, responsive AI chatbot website using React 18, Vite, and Tailwind CSS for fast performance and a seamless user experience.
- Integrated dynamic routing with React Router and modular architecture with reusable components for scalability.
- Deployed backend chatbot service alongside a styled frontend based on community Figma designs, enabling user-friendly conversational AI access.